

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”

Problem Statement -

Problem Statement : Interview Trainer Agent

The Challenge

Students and **job seekers** often struggle with interview preparation, question selection, and performance evaluation. Traditional preparation methods provide generic questions and lack personalized practice, realistic interview simulations, and actionable feedback. Without intelligent guidance, candidates may find it difficult to identify their weak areas and improve their **interview performance**.

The Objective

Build an **AI-powered Interview Agent** that delivers personalized interview preparation, realistic practice, and performance feedback for candidates. The solution should:

Personalized Interview Preparation – Generate questions based on the candidate's resume, skills, and target role.

AI Question Generation – Create relevant technical, HR, and behavioral interview questions.

Interview Simulation – Conduct structured mock interviews to provide a realistic practice experience.

Answer Evaluation & Feedback – Analyze responses and provide constructive feedback and improvement suggestions.

Performance Analysis – Identify strengths, weaknesses, and areas requiring improvement.

Multi-Agent System : –

Interview Question Agent – Generates relevant interview questions.

Interview Evaluation Agent – Evaluates candidate responses.

Feedback Agent – Provides personalized feedback and recommendations.

Performance Dashboard – Display interview results, scores, strengths, weaknesses, and overall progress.

Project Tile – InterviewIQ-Agent

Domain of Project – Education & Career Development

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
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Recommended font style and size:

Font Style: Arial

Font Size (Heading): 28

Font Size (Content): 20

Proposed Solution -

Proposed Solution: AI-Powered InterviewIQ-Agent

The proposed solution is a **multi-agent AI-powered Interview Agent** built using **IBM BOB**, **IBM watsonx.ai**, and **IBM Granite Models** to provide **personalized interview preparation**, realistic **mock interviews**, and **intelligent performance feedback**.

The solution uses AI to analyze the candidate's resume, skills, target role, and interview responses to generate relevant questions and evaluate performance.

The system consists of key components:

Interview Question Agent – Generates personalized technical, HR, and behavioral questions.

Interview Simulation Agent – Conducts structured mock interview sessions.

Answer Evaluation Agent – Analyzes responses and evaluates relevance, clarity, and quality.

Feedback Agent – Provides personalized feedback and improvement suggestions.

Performance Dashboard – Displays scores, strengths, weaknesses, and overall interview progress.

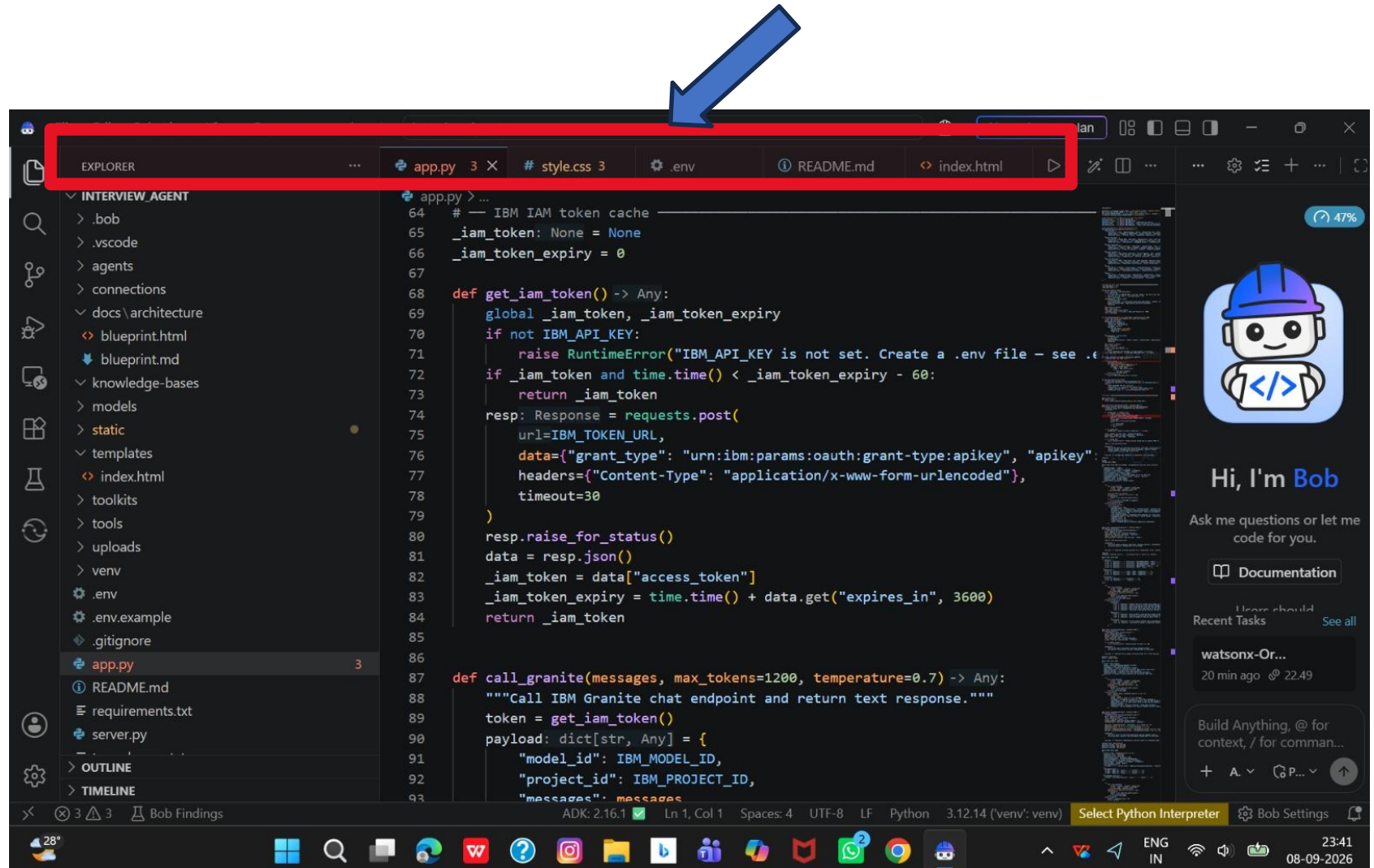
The solution provides an **end-to-end** intelligent interview preparation platform that helps candidates practice effectively, identify skill gaps, and improve their interview readiness through **AI-driven guidance** and **feedback**.

Technology Used

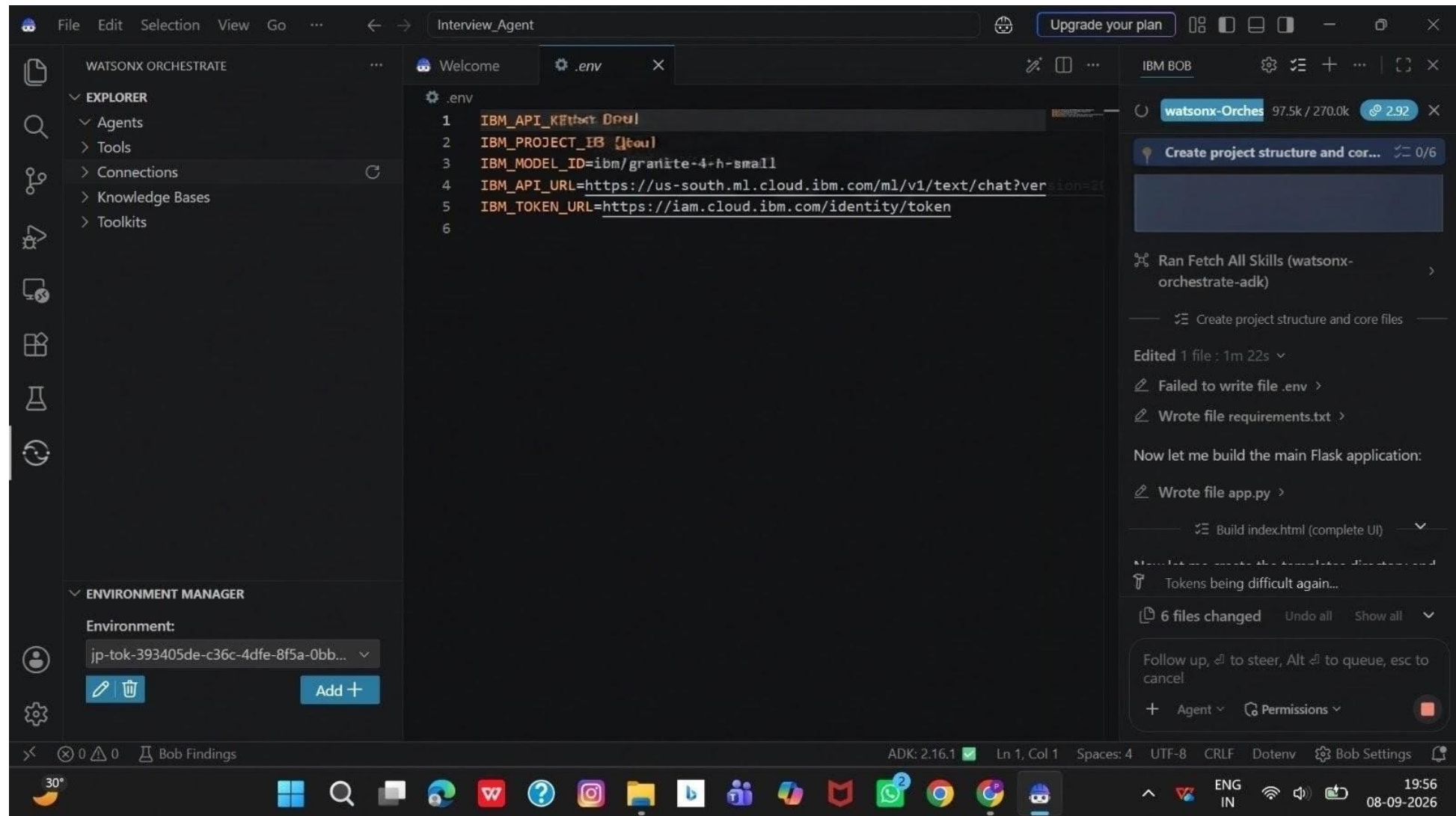
- **IBM Bob platform** -For designing and orchestrating multi-agent interview workflows visually
- **IBM Granite model** - for eg – ibm-granite-3-2-8b – For natural language understanding, reasoning,question generation and personalized interview feedback.
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the InterviewIQ Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the InterviewIQ Agent.
- **RAG–** RAG to **fetch real data first, then generate intelligent answers**, Retrives relevant information from the candidate's resume and knowledge sources to generate accurate, personalized interview questions
- **IBM Watsonx.ai** - For model deployment, AI development, embeddings, and AI governance
- **Vector Database** – For efficient RAG-based retrieval of resume, skills, and interview-related information

Required Files created using IBM bob -

1. App.py
2. Requirements.txt
3. Style.css
4. .env
5. Index.html



IBM Bob Screenshots:



File Edit Selection View Go ...

Interview_Agent

Upgrade your plan

WATSONX ORCHESTRATE

EXPLOLER

Agents

Tools

Connections

Knowledge Bases

Toolkits

ENVIRONMENT MANAGER

Environment:

jp-tok-393405de-c36c-4dfe-8f5a-0bb...

Add +

Welcome

Bob Edit Preview of deploy_ngrok.py

C: > Users > teja8 > OneDrive > Desktop > Interview_Agent > deploy_ngrok.py

```
1+ """
2+ InterviewIQ Public deployment launcher
3+ Uses pyngrok to tunnel the local waitress server to a public HT
4+ Run: python deploy_ngrok.py
5+ """
6+ import os
7+ import sys
8+ import time
9+ import threading
10+ import webbrowser
11+
12+ # -- Load env --
13+ from dotenv import load_dotenv
14+ load_dotenv()
15+
16+ PORT = int(os.getenv("PORT", 5000))
17+ NGROK_TOKEN = os.getenv("NGROK_AUTHTOKEN", "")
18+
19+ # -- pyngrok --
20+ try:
21+     from pyngrok import ngrok, conf
22+ except ImportError:
23+     print("ERROR: pyngrok not installed. Run: pip install pyngro
24+     sys.exit(1)
25+
26+ # -- waitress --
27+ try:
28+     from waitress import serve
```

IBM BOB

watsonx-Orche 109.3k / 270.0k 6.44

Create project structure and cor... 0/6

Let me first inspect the project thoroughly before attempting any deployment.

Listed files in . >

Ran 5 commands >

Good — pyngrok and waitress (Windows-compatible WSGI server) are available. Now let me check if there's an ngrok auth token available and inspect the app's current state:

Tools awaiting approval:

Write File deploy_ngrok.py Open Diff

Approve once Reject

Approve edit tools for task

6 files changed Undo all Show all

Follow up, ⌘ to steer, Alt ⌘ to queue, esc to cancel

Agent Permissions

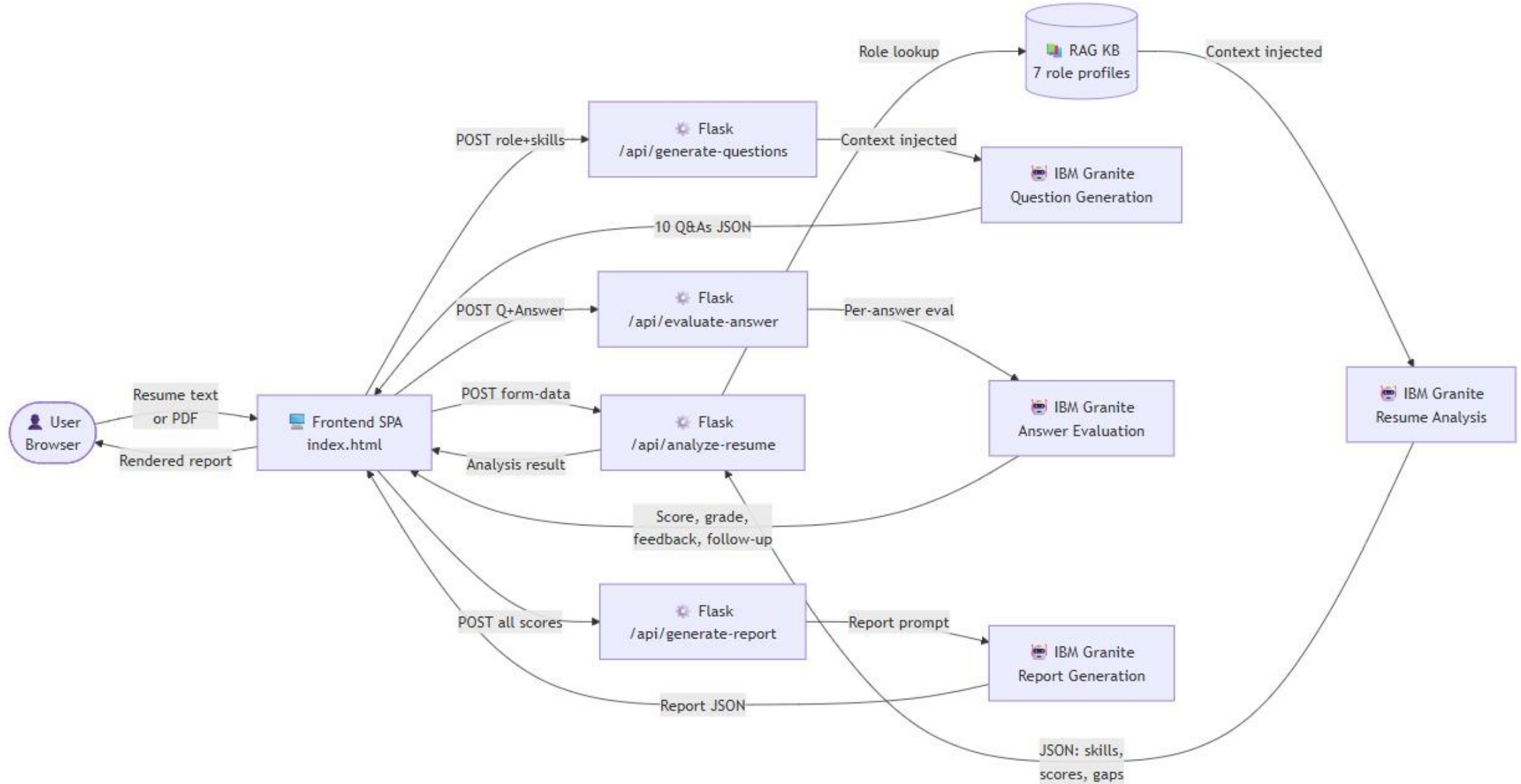
ADK: 2.16.1 Ln 1, Col 1 Spaces: 4 UTF-8 LF Python 3.12.14 ('venv': venv) Bob Settings

29°

Windows Taskbar

20:35 08-09-2026

Architecture blueprint for IBM bob:



Role of Agentic AI in the solution

Role of Agentic AI in InterviewIQ Agent

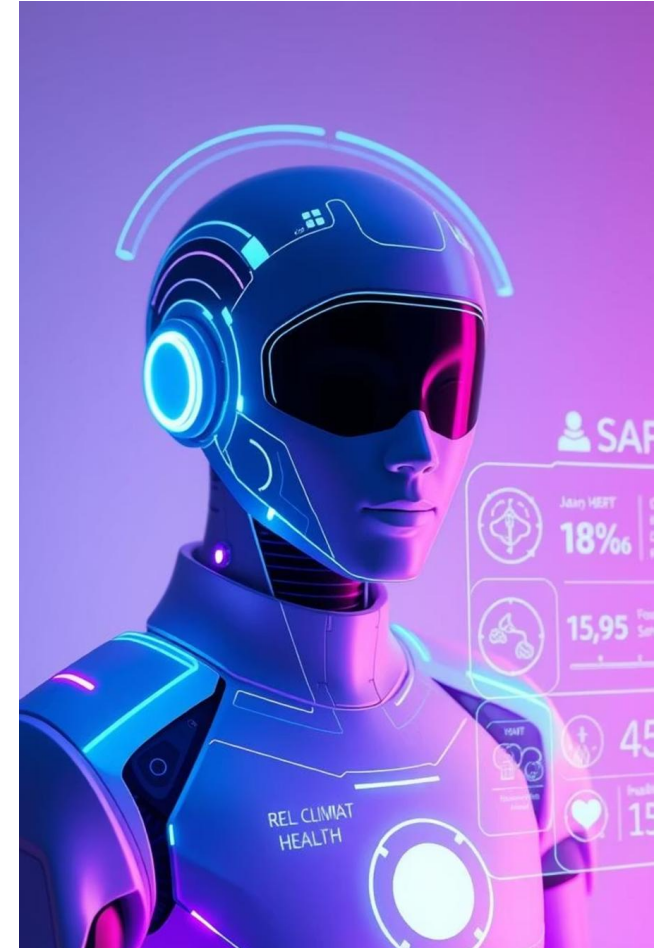
Agentic AI enables the **InterviewIQ Agent** to function as an **intelligent, autonomous** interview-preparation system rather than a simple chatbot. Using **IBM watsonx.ai**, **IBM Granite models**, and **IBM Bob**, it coordinates multiple specialized agents to provide personalized interview preparation.

Each agent performs a specific role—such as analyzing the user's **resume**, **retrieving** relevant **interview questions**, generating **technical** and **HR questions**, **evaluating answers**, and **providing feedback**—while working collaboratively. This multi-agent approach ensures efficient task execution and relevant results.

Agentic AI brings context awareness by understanding the user's **skills**, **resume**, **target role**, and previous responses to generate tailored interview questions and recommendations. It supports **real-time adaptation**, continuously adjusting **questions** and **feedback** based on the user's **answers** and **performance**.

Additionally, agents communicate with each other to improve **decision-making**, question relevance, answer evaluation, and personalized feedback.

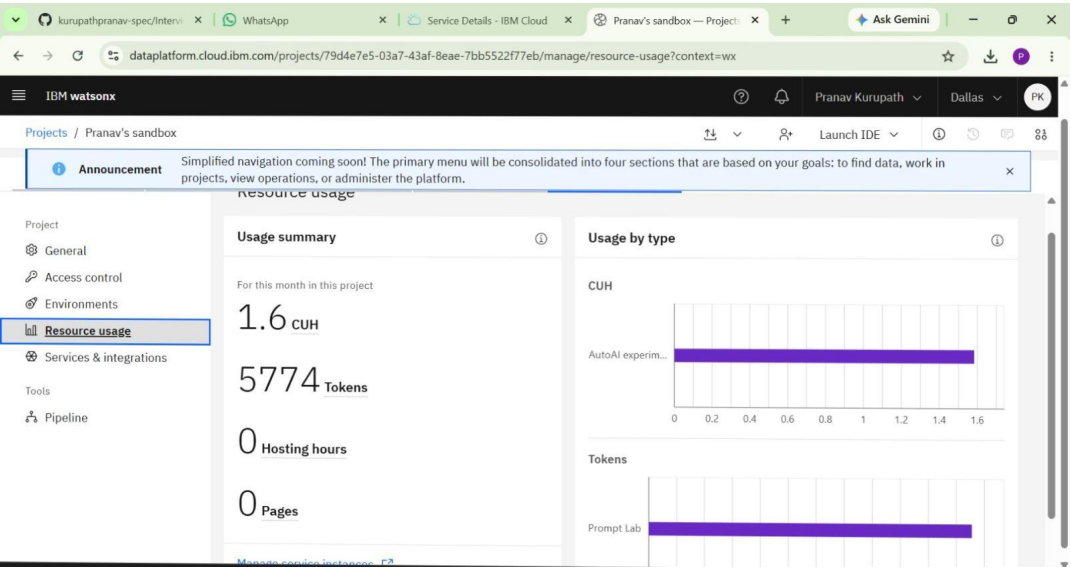
Overall, Agentic AI makes **InterviewIQ Agent** proactive, personalized, adaptive, and **goal-driven**, acting as a smart digital interview coach that helps users improve their interview readiness.



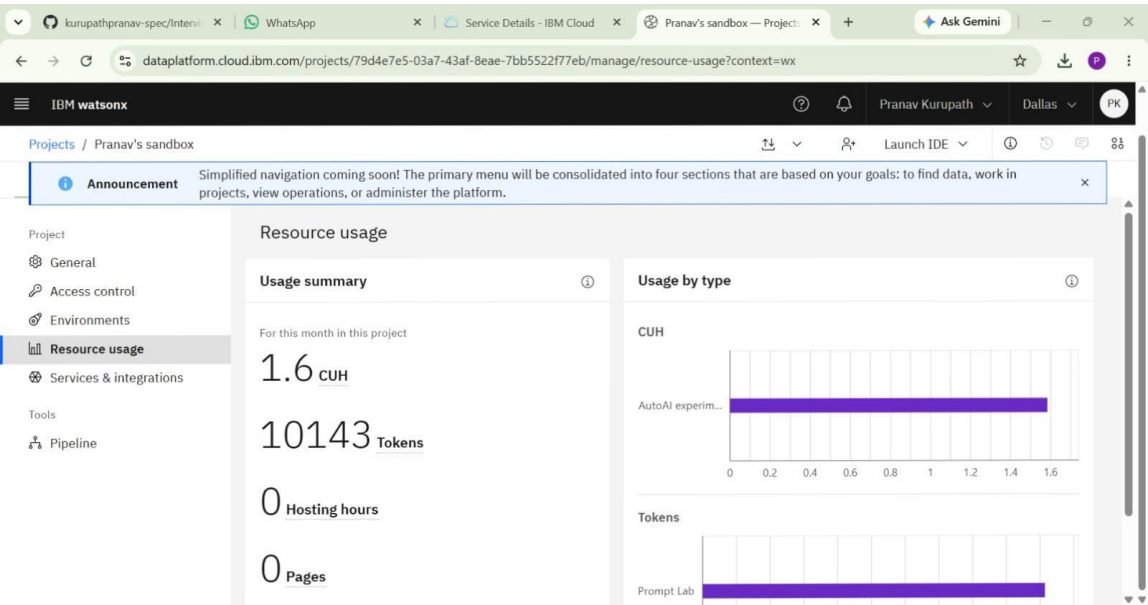
Token Usage

4369 token are used

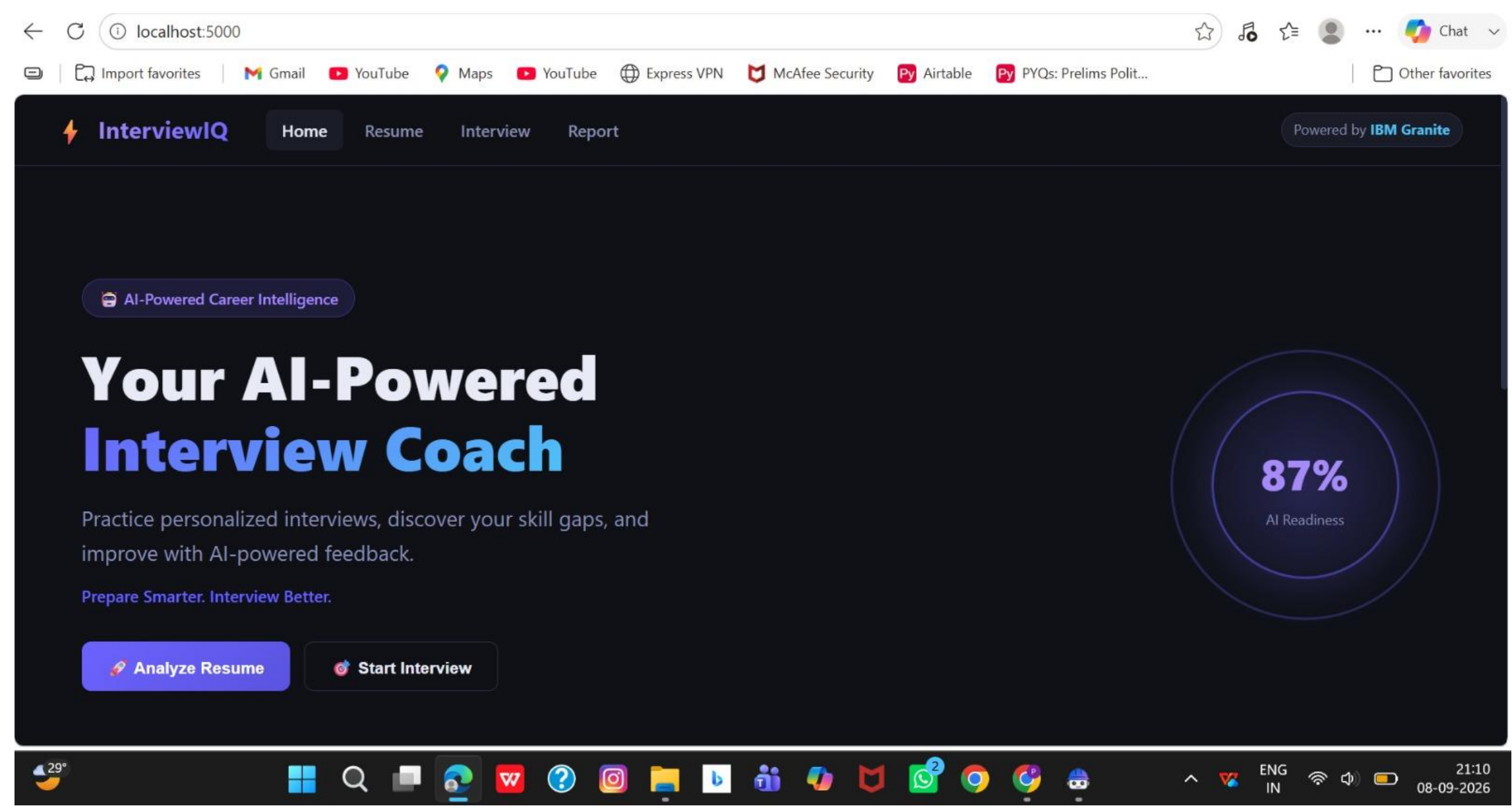
Before Prompt



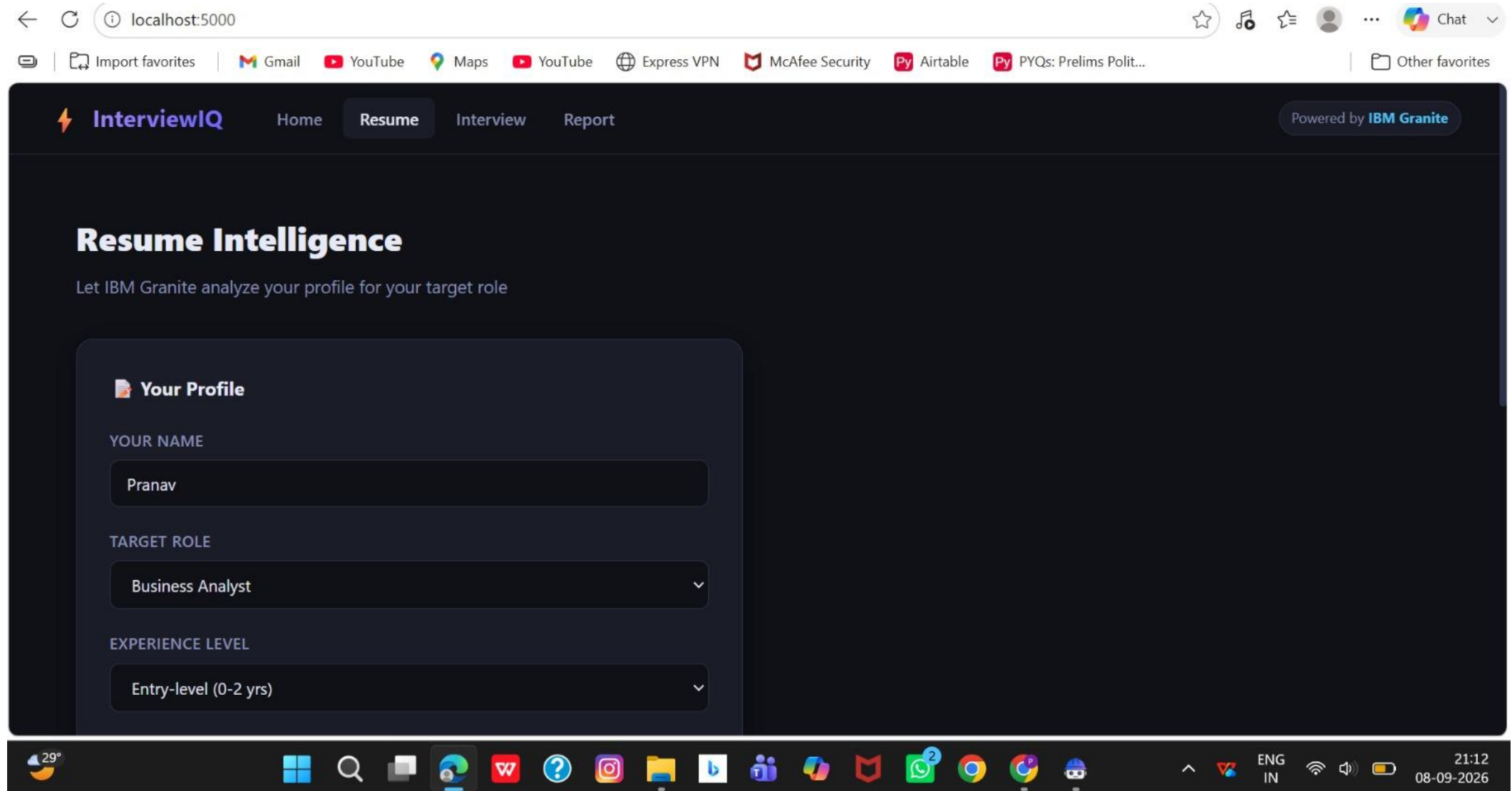
After the Prompt



Project Output Screenshot



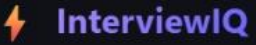
Project Output Screenshot



Project Output Screenshot

← ↻ ⓘ localhost:5000 ☆ 🎧 ⋮ 👤 ... 🗨 Chat ▾

📁 Import favorites | 📧 Gmail | 📺 YouTube | 📍 Maps | 📺 YouTube | 🌐 Express VPN | 🛡 McAfee Security | 📌 Airtable | 📌 PYQs: Prelims Polit... | 📁 Other favorites

 InterviewIQ

Home Resume Interview Report

Powered by IBM Granite

Resume Intelligence

Let IBM Granite analyze your profile for your target role

Your Profile

YOUR NAME

Pranav

TARGET ROLE

Business Analyst ▾

EXPERIENCE LEVEL

Entry-level (0-2 yrs) ▾

INTERVIEW FOCUS

Behavioral ▾

Analysis Results

65%
Readiness

80%
Technical

75%
Communication

68%
Role Fit

✅ DETECTED SKILLS

Python

SQL

Power BI

Excel

Data Modelling

ETL

KPI Reporting

Power Query

Data Analysis

Dashboard Development

Generative AI

LLMs

Prompt Engineering

GitHub

Google Colab

Critical Thinking

Quick Learning

Self-Motivated

Problem Solving

Team Collaboration

Communication

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Novelty and Uniqueness

The proposed **InterviewIQ** Agent stands out by combining **Agentic AI**, **RAG**, and personalization into a single intelligent **interview-preparation system**.

Multi-Agent Intelligence – Multiple specialized agents collaborate to provide personalized interview preparation.

RAG-Based Accuracy – Retrieves relevant interview information and combines it with AI generation for reliable questions and guidance.

Hyper-Personalization – Tailors interview questions based on the user's resume, skills, experience, and target job role.

Real-Time Feedback Loop – Continuously adapts questions and recommendations based on the user's answers and performance.

Multi-Modal Interaction – Supports text-based interaction and can be extended to voice-based interview practice.

Technical + HR Preparation – Covers technical, coding, behavioral, and HR interview questions in one platform.

Performance Analysis – Evaluates responses and provides actionable feedback to help users improve.

Scalable & Enterprise-Ready – Built using IBM watsonx.ai, IBM Granite models, and IBM Bob, supporting reliable AI workflows and deployment.

This makes InterviewIQ Agent a **smart, adaptive, and proactive digital interview coach**, not just a static question-generation tool.

Git Hub Link

- Please create public github repository (Enable Add README button while creating repository)

with format –

Please test and Paste the working Github URL – [Github](#)

Make sure you have Uploaded following three files into your github repository

1) All Agent relevant files of IBM Cloud , 2) yourproblemstatement.pdf file 3) your projectpresntation.pptx file 4)any other files

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Future Scope

1. **AI-Powered Mock Interviews** – Add real-time mock interview sessions where the system asks questions, evaluates responses, and provides personalized feedback.
2. **Voice-Based Interview Practice** – Enable voice interaction for a more realistic interview experience, including speech-based answers and feedback.
3. **Multilingual Support** – Support multiple languages to make interview preparation accessible to a wider range of users.
4. **Advanced Performance Analytics** – Provide detailed dashboards showing strengths, weaknesses, improvement areas, and interview readiness scores.
5. **Job-Specific Preparation** – Integrate job descriptions to automatically generate role-specific questions and preparation plans based on required skills.
6. **Continuous Learning** – Improve recommendations over time by learning from previous interview sessions, user performance, and feedback.

Certificates Earned:

Please add your credly certificate(Getting Started with Artificial Intelligence)



Certificates Earned:

Add your IBM BOB Certificate

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to
Pranav Kurupath

for the completion of

**Lab: Troubleshoot Your
Code Using IBM Bob**

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 31 Aug 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.